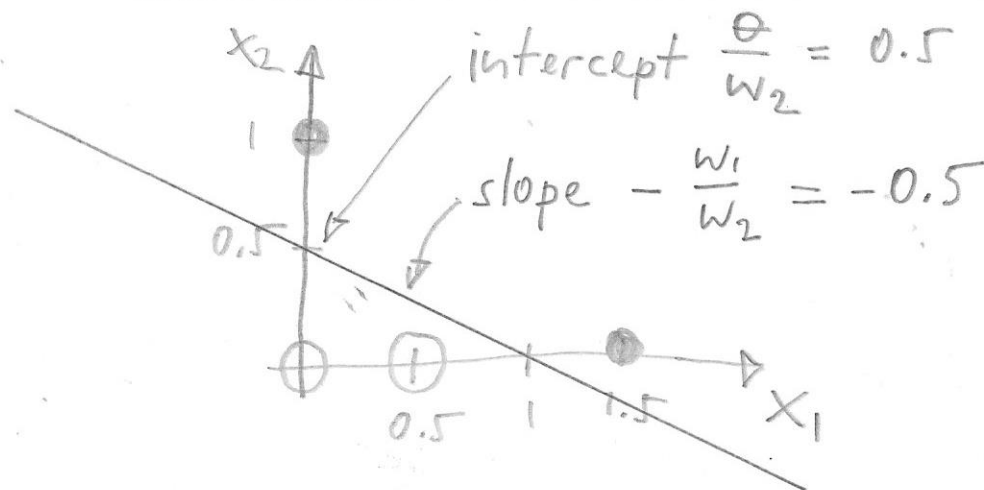


Exercise 1: simple perceptron

(1) Draw the input space for the following classification:

x_1	x_2	target
0	0	0
0.5	0	0
0	1	1
1.5	0	1

(2) Draw a possible decision line in input space. What is the equation describing this line, and what are possible weights and a threshold that would result in such a decision line?



possible solution:

$$\theta = 1$$

$$w_1 = 1$$

$$w_2 = 2$$

$$x_2 = -0.5 \cdot x_1 + 0.5$$